

Test 1 corrections

Q3 a) $\frac{18!}{18} = 17!$ linear permutation
 $\uparrow$
 k ways to pick a start of group

b) $1 \cdot 15 \cdot 16!$ $= 16! \cdot 15$
 $\uparrow$ $\uparrow$ $\uparrow$
 # ways to sit at A # ways to sit at B k ways to sit everyone else

Q4 Consider $t-1$ people, make everyone not know each other. Then there is a group of $t-1$, all who don't know each other. Since no one knows each other and the group is $t-1 < t$, so:

$$K_{t-1} \not\rightarrow K_2, K_n$$

Therefore

$$R(3, t) \geq t$$



Q5 b) for all permutations of $\{1, 2, \dots, n\}$, insert $n+1$'s in between

$$n! \cdot \binom{n+1}{n} = n! \cdot \binom{n+1}{1} = n! \cdot (n+1) = (n+1)!$$

$\uparrow$
pick gaps

c) $\frac{(2n-2)!}{n!} \cdot (2n-1) \cdot 2! = \frac{2 \cdot (2n-1)!}{n!}$

place all except 1 and 2

insert pair $\uparrow$
 $\uparrow$
 (12) or (21)